Victoria K. Makowski

NSF RaMP Research Fellow
email: vmakowski@outlook.com || website: marshwranglers.com

Education

Augusta University (AU) Augusta, GA
B.S. in Ecology; *Magna Cum Laude; Distinction of Excellence in Research* May 2025

- GPA: 3.71 Institutional / 3.72 Overall
- Related Coursework: Statistics, Physics I & II, General Chemistry I & II, Organic Chemistry I, Ecology, Genetics, Community Field Ecology, Evolutionary Biology, Marine Pollution, Ornithology, Zoology.

Technical Skills

- **Field Skills:** Electrofishing, Cameras, Nets (Cast, Dip, Experimental Gill, Cotton & Monofilament Seine, & Trawl), Species Identification & Transport, Surber Samplers, Traps (Crab, Eel, & Minnow), Water Quality Monitoring, Greenhouse Systems, Large Vehicle & UTV Driving.
- **Lab Skills:** Data Recording & Processing, Dissection, Fish Husbandry, Herbarium/Museum Care, Mesocosm Creation, Micro & Macroscopes, Organism Preservation & Documentation.
- **Computer Skills:** 3D Printing, Arduino Programming, C++ Basics, CAD Design, Google Suite, Microsoft Suite, RStudio Basics, Python Basics.

Licenses & Certifications

- AU & GS Biology Department & University Vehicle Certification
- AU Biology Department Certified Field Assistant & Mentor
- CITI Certifications: Responsible Conduct of Research; Animal Researcher; Wildlife Researcher; Working with Fish in Research Settings; Animal Biosafety; Emergency and Incident Response; Lab Chemical Safety

Affiliations & Honor Societies

- Society for the Preservation of Natural History (SPNHC) 2025 – *Present*
- Coastal & Estuarine Research Federation (CERF) 2025 – *Present*
- Southeastern Estuarine Research Society (SEERS) 2023 – *Present*
- Tri Beta Science Honor Society 2023 – *Present*
- AU Biology Club: *Vice President* (2024 – 2025) 2021 – 2025

Awarded Grants & Honors

- Most Outstanding Senior in Biology 2025
- SEERS Student Travel Grant 2025
- AU Center for Undergraduate Research (CURs) Student Research Grant 2025
- NSF's Promoting Opportunities & Pathways for Undergraduate Persistence in STEM (POPUPS) 2022 – 2025

Professional History

Georgia Southern University (GS) Department of Biology Statesboro, GA
Research and Mentoring for Postbaccalaureates (RaMP) Research Fellow 2026 – *Present*

- NSF-funded Mentoring and Research Opportunities for Careers in Coastal Science (MROC²S).
- Advisor: Dr. Johanne Lewis
- Project: Burning Hearts: Cardiovascular thermal tolerance of hybrid vs. striped bass
- Assessing fish cardiovascular stress in relation to increased water temperature.

AU Department of Physics & Biophysics Augusta, GA
Project Lead; Augusta Utilities Research Intern 2024 – 2026

- Advisor: Dr. Joseph Hauger
- Project: DOpTC: Autonomous Water Quality Sensing
- Engineered low-cost water sensors for Augusta Utilities' Smart City initiative & the Satilla Riverkeeper.
- Worked in a small group creating a sonde measuring DO, ORP, pH, Temperature, & Conductivity.
- Calibrated and coded Atlas Scientific probes. Designed the housing in CAD and 3D-printed it.
- Two versions: one stationary buoy to leave in the environment, one mounted on a kids' kayak with both autonomous & remote-control modes.

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AU Department of Biological Sciences Augusta, GA

Boat Lead; Undergraduate Research Assistant 2025 – *Present*

- Advisor: Drs. Chris Bates, Stacy Bennetts, Joseph Hauger, Loren Mathews, & Jessica Reichmuth
- Project: Monitoring Local, State, & Federal Restoration Efforts of an Obsolete Navigational Channel
- A collaborative tri-university project assessed the ecological impacts of cut closures on the Satilla River.
- Top-down forces (fish/mobile invertebrate diversity, crustacean population structure, & food web analysis) were monitored using boat-deployed experimental gill nets, trawl nets, & counting crab burrows.
- Bottom-up parameters (water chemistry, light availability, phytoplankton abundance, & diversity of salt marsh plants & sediment microbes) were monitored using sonde readings, marsh transects, & soil analyses.

Undergraduate Research Assistant 2025 – 2026

- Advisor: Drs. Loren Mathews & Jessica Reichmuth
- Project: One fish, two fish, *more* fish? Are channel closures positively impacting Satilla River fish assemblages?
- Analyzed 6+ years of fish assemblage data with Shannon Diversity & cross-referenced with water salinity, turbidity, & temperature before & after Satilla River cut closures.

Undergraduate Field Assistant 2024

- Advisor: Dr. Randy Singer
- Project: Fish Slam
- Worked with USGS, Florida DNR, & Florida FWC collecting non-native species around Southern Florida.
- Freshwater & brackish systems were sampled for non-native fishes using electrofishing boats, hook-and- line, minnow traps, dip nets, & cast nets. Non-native fishes were ethically euthanized.

Independent Researcher 2024 – 2025

- Advisor: Dr. Stacy Bennetts
- Project: Chloroplasts From the Past: Digitizing AU's Herbarium
- 1000+ herbarium specimens were organized & cataloged into a digital platform to be uploaded to a widely accessible online database, *Symbiota*.

Independent Researcher 2024 – 2025

- Advisor: Dr. Jessica Reichmuth
- Project: Specs of Speckled Crabs: Observations of why speckled crabs may not be good lab specimens
- Speckled crabs were collected by hand or with bag & monofilament seine nets of various lengths.
- Sex, coloration, & carapace width were measured & compared to after-molt data.

Lead Greenhouse Technician 2024 – 2025

- Maintained & improved the two greenhouses located on the AU campus.
- Gained proficiency in greenhouse climate systems and plant husbandry.

Coursework Researcher 2023

- Advisor: Dr. Jessica Reichmuth
- Project: Comparing microplastic tolerance & behaviors of male sand fiddler crabs, *Uca pugilator*, from Hunting Island & Tybee Island
- Analyzed fiddler crab tolerance & behavior changes to dietary microplastics between different anthropogenically influenced locations.
- Collected crabs by hand & proficiently recognized SC & GA invertebrate salt marsh genera.

Lead Undergraduate Researcher 2023 – *Present*

- Advisor: Dr. Jessica Reichmuth
- Project: Oceanic fish diversity among barrier islands along the Georgia coast
- Utilized bag & monofilament beach seines, minnow traps & crab pots in fresh & brackish water.
- Identified fish to species using field guides, counted & measured all fish caught.
- Collected, transported, & acclimated live specimens into the AU aquarium.

Presentations & Publications

- **Makowski, V.,** Byne, W., Flite, O., Hauger, A., Herndon, E., Lau, N., Leavins, M., & Reichmuth, J. (2026). *Stream-ing Success: Engineering a Low-Cost, Remote Water Monitoring System.* [Publication; not peer-reviewed] The Georgia Operator, Summer 2026 Issue.

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- Leavins, M., Herndon, E., Lau, N., **Makowski, V.,** Hauger, J., & Reichmuth, J. (2026, July 31). *DOpTC: Low-Cost Water Quality Monitoring* [Oral presentation]. Augusta Utilities 2026 End of Summer Research Presentations, Augusta, GA, United States.
- Leavins, M., Herndon, E., Lau, N., **Makowski, V.,** Hauger, J., & Reichmuth, J. (2026, March 22 - 26). (2026, April 17) *DOpTC: A Low-Cost Remote Water Quality Monitoring System.* [Poster session]. ACS 2026 Spring Meeting, Atlanta, GA, United States. AU URFA 2026 Conference, Augusta, GA, United States. https://www.acs.org/events/acs-meetings/past-meetings.html https://www.augusta.edu/curs/urfa/
- **Makowski, V.,** Herndon, E., & Reichmuth, J. (2025, November 9 - 13). *One fish, two fish, more fish? Are channel closures positively impacting Satilla River fish assemblages?* [Poster session]. CERF 2025 Biennial Conference, Richmond, VA, United States. https://conference.cerf.science/
- Herndon, E., Lau, N., Leavins, M., **Makowski, V.,** Hauger, J., & Reichmuth, J. (2025, November 4 - 5). (2026, February 7). *Low-Cost Remote Water Quality Monitoring (DOpTC).* [Oral presentation]. GAWP 2025 Fall Conference & Expo, Athens, GA, United States. Satilla Science Dover Bluff Community Update & Outreach. https://www.gawp.org/events/EventDetails.aspx?id=1994419
- Herndon, E., Lau, N., Leavins, M., **Makowski, V.,** Hauger, J., & Reichmuth, J. (2025, July 30). *DOpTC: Low-Cost Water Quality Monitoring* [Oral presentation]. Augusta Utilities 2025 End of Summer Research Presentations, Augusta, GA, United States.
- **Makowski, V.** & Bennetts, S. (2025, April 8). (2025, May 27 - 30). *Chlorolasts From the Past: Digitizing Augusta University's Herbarium.* [Poster session]. AU URFA 2025 Conference, Augusta, GA, United States. SPNHC 2025 Convention, Lawrence, KS, United States. https://www.augusta.edu/curs/urfa/index.php https://spnhc2025.ku.edu/
- Cooke, W., Herndon, E., Leavins, M., **Makowski, V.,** Wilcher, A., Byne, W., & Flite, O., Hauger, J., & Reichmuth, J. (2025, April 8). *DOpTC: A Low-Cost Water Quality Monitoring System.* [Oral presentation]. AU URFA 2025 Conference, Augusta, GA, United States. https://www.augusta.edu/curs/urfa/index.php
- **Makowski, V.,** & Reichmuth, J. (2025, March 6-8). *Specs of Speckled Crabs: Observations of why speckled crabs may not be good laboratory specimens.* [Poster session]. SEERS 2025 Convention, Fort Pierce, FL, United States. https://seers.org/meeting-history/2023-spring-2-2/
- Herndon, E., Leavins, M., **Makowski, V.,** Wilcher, A., Hauger, J., & Reichmuth, J. (2024, August 16). *Remote Water Quality Monitoring* [Oral presentation]. Augusta Utilities 2024 End of Summer Research Presentations, Augusta, GA, United States.
- Blumling, E., Ekechukwu, J., **Makowski, V.,** & Reichmuth, J. (2024, March 21 - 23). *Comparing microplastic tolerance & behaviors of male sand fiddler crabs, Uca pugilator, from Hunting Island & Tybee Island: The Crustacean Weight Watchers.* [Poster session]. SEERS 2024 Convention, Savannah, GA, United States. https://seers.org/meeting-history/2023-spring-2/

References

Dr. Jessica Reichmuth, PhD.
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